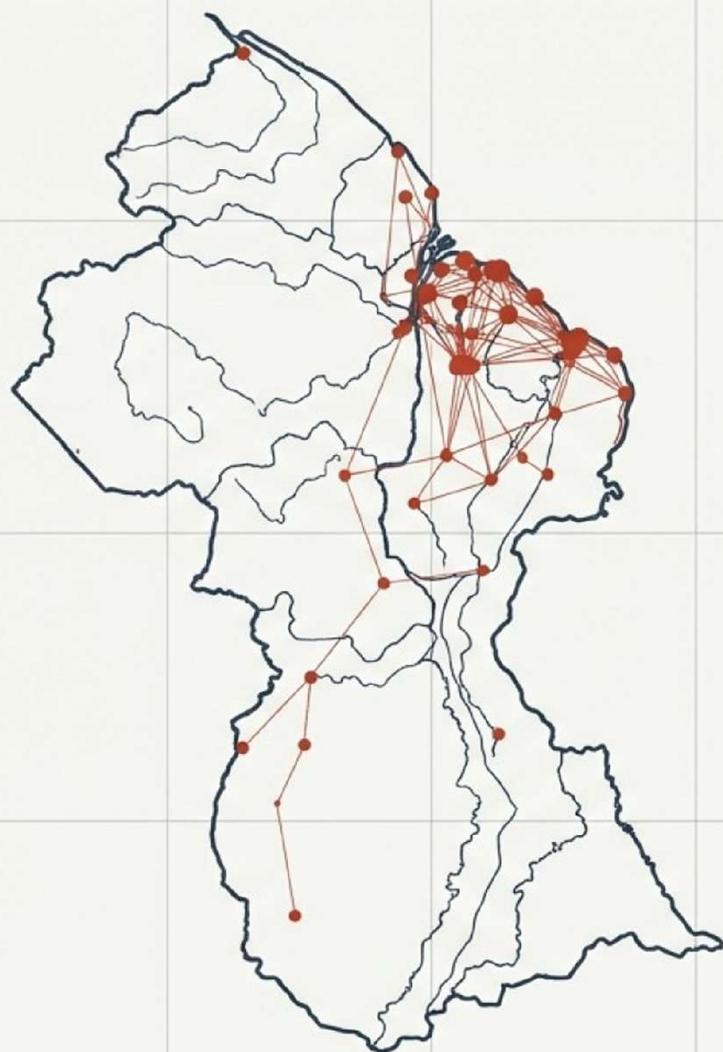




Explaining the Limited Continued Use of the GRA “Padna”App

A Quantitative Study on E-Government Continuance in Guyana

FORMAL RESEARCH PROPOSAL | FYRP 4100

Gaianand Ramnarace - 1045379

Digital Transformation: The Gap Between Access and Sustained Use

The Context

Governments digitize to improve efficiency and transparency (UN, 2022).

Guyana is undergoing rapid digital transformation to reform the public sector.

The Guyana Revenue Authority (GRA) launched the 'Padna' App to streamline tax transactions including filing, compliance, and payments.



'Padna' App

The Reality & Stakes

Availability does not guarantee long-term engagement.

E-government services often experience a sharp drop in usage after launch. Without continuance (sustained use), the ROI of digital platforms is lost, and citizens return to paper-based lines and in-person processing.



Paper-Based Queue

Literature Review

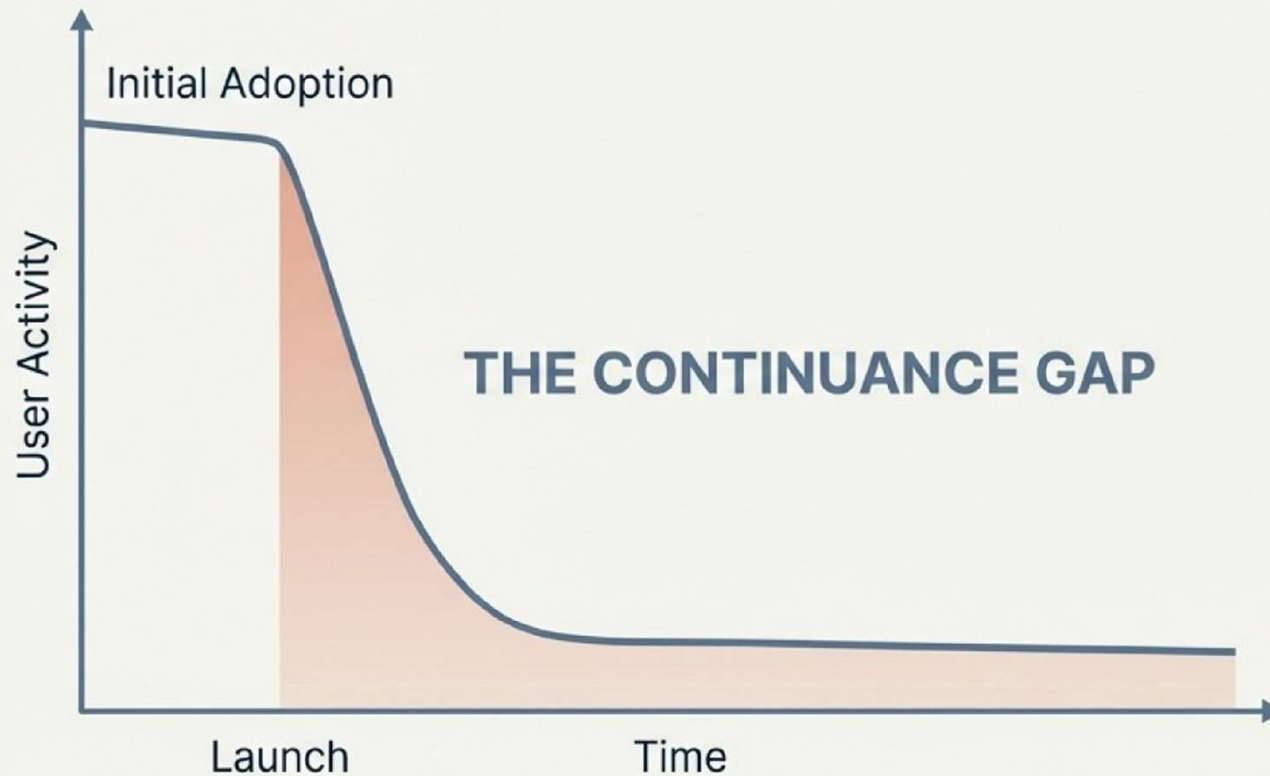


-“Post-adoption research indicates that attrition can occur even when systems are functionally adequate, because continued use depends on perceived value, user experience and contextual constraints rather than technical availability alone (Bhattacharjee, 2001; Haug et al., 2023).”

-“Studies that segment users by residential status or infrastructural context demonstrate that these constraints can materially change relationships among service quality, satisfaction and continued use, suggesting that models should incorporate enabling conditions when research settings exhibit pronounced inequalities (Kala et al., 2024).”

-“Recent continuance evidence shows trust operates both directly and indirectly. Luo et al. (2024) position trust as a mediator through which confirmation and usefulness can translate into continued use, suggesting that sustained engagement requires not only perceived benefit but also confidence in the platform’s reliability and fairness.”

The 'Continuance Gap' in the Guyanese Tax System



The Core Problem:

Citizens adopt the Padna App but often discontinue use, returning to traditional non-digital channels.

The Barriers:

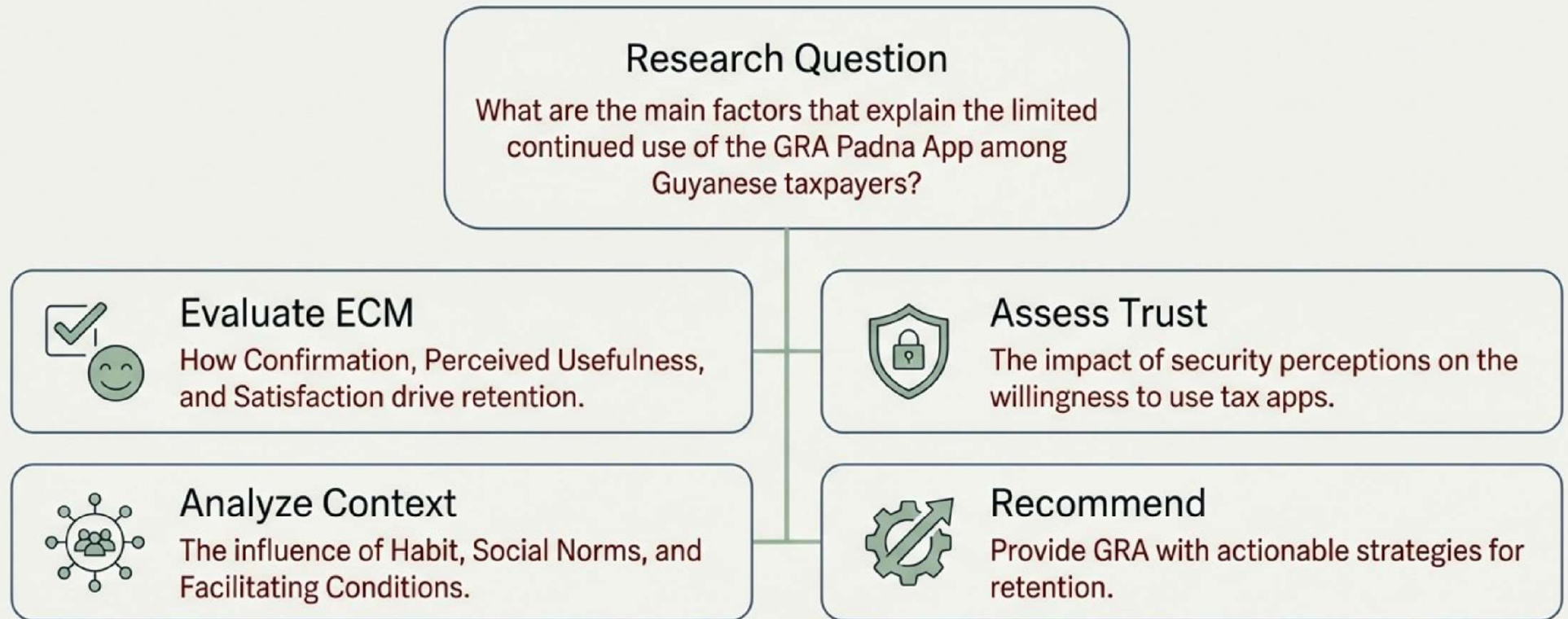
- **Structural:** Uneven connectivity (Hinterland vs. Coast) (UNDP, 2023).
- **Social:** Persistent trust and privacy concerns regarding financial data.
- **Experience:** Technical issues and variable digital literacy.

The Evidence Gap:

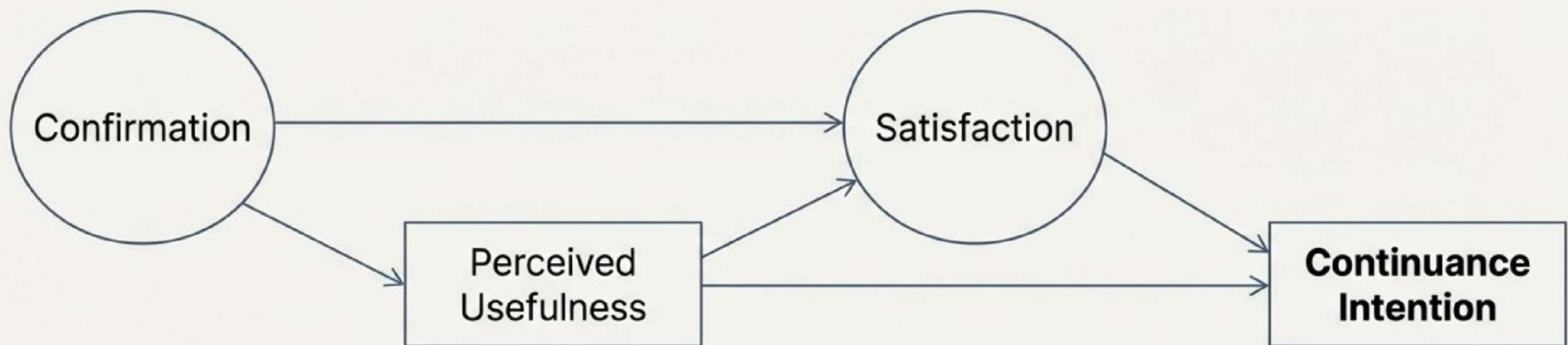
No empirical studies currently explain the specific determinants of discontinuance for the Padna App.



Defining the Investigation Scope



Theoretical Lens: The Expectation-Confirmation Model (ECM)



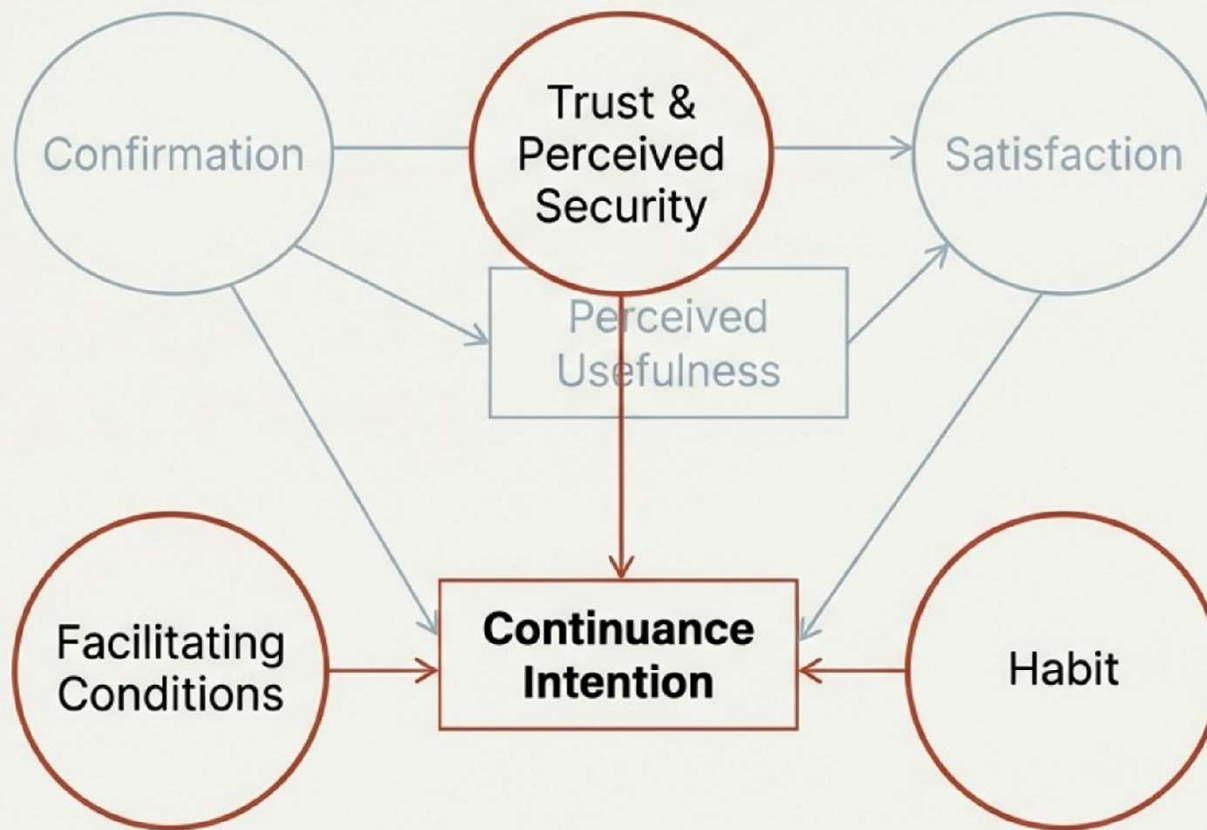
Why ECM?

The influential framework for explaining post-adoption behavior (Bhattacharjee, 2001). Adoption theories focus on the 'first time'; ECM explains the 'next time'.

Critique

ECM is robust but insufficient on its own for the complex, mandatory nature of government services in developing contexts (Luo et al., 2024).

Adapting Theory to Context: An Integrated Research Model



Trust & Security:

Essential for tax environments. Low trust leads to immediate discontinuance (Bélanger & Carter, 2008).

Facilitating Conditions:

Addresses the 'Digital Divide' (infrastructure/literacy) prominent in Guyana (UNDP, 2023).

Habit:

Addresses the infrequency of tax cycles; use must be routinized to survive long gaps (Venkatesh et al., 2012).



Research Design & Methodology



Philosophy

Positivist / Deductive.
Testing established
theories against empirical
data.



Design & Instrument

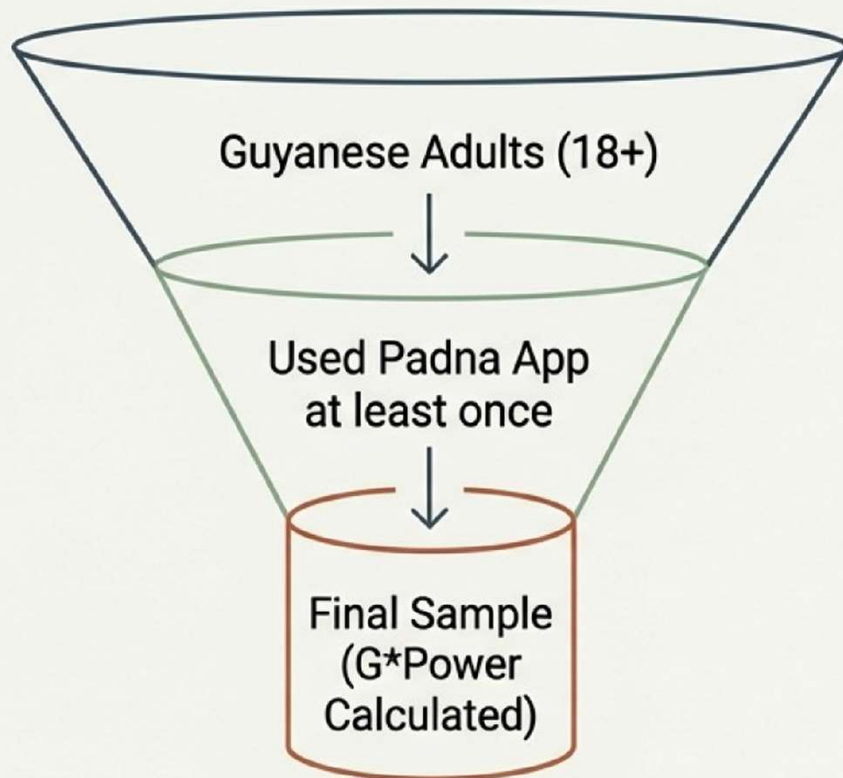
Cross-sectional
Explanatory Survey.
Online questionnaire
using validated Likert-type
scales (adapted from
Bhattacharjee, 2001;
Venkatesh et al., 2012).



Data Analysis

Partial Least Squares
Structural Equation
Modelling (PLS-SEM)
using SmartPLS 4.
Ideal for prediction-
oriented modeling with
complex latent constructs.

Target Population & Sampling Strategy



Sampling Method

Non-probability purposive sampling + broad online distribution (social media, professional networks).



Inclusion Criteria

Residents of Guyana, 18+, prior use of Padna. 'Continuance' can only be measured after an initial experience.

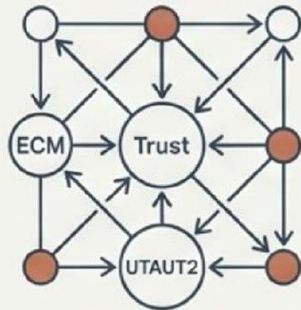


Ethical Safety

Anonymous data collection; no Taxpayer Identification Numbers (TINs) recorded.

Significance of the Study

Academic Contribution



Validates an integrated model (ECM + Trust + UTAUT2) in a Caribbean context.

Addresses the "Context Gap" in IS research (currently concentrated in Asia/Middle East).

Policy & Industry



Provides GRA with evidence to improve UX, service reliability, and support structures.

Identifies specific "pain points" causing user churn.

Societal Impact



Reduces administrative burdens on citizens.

Strengthens confidence in digital public systems, supporting broader digital inclusion.

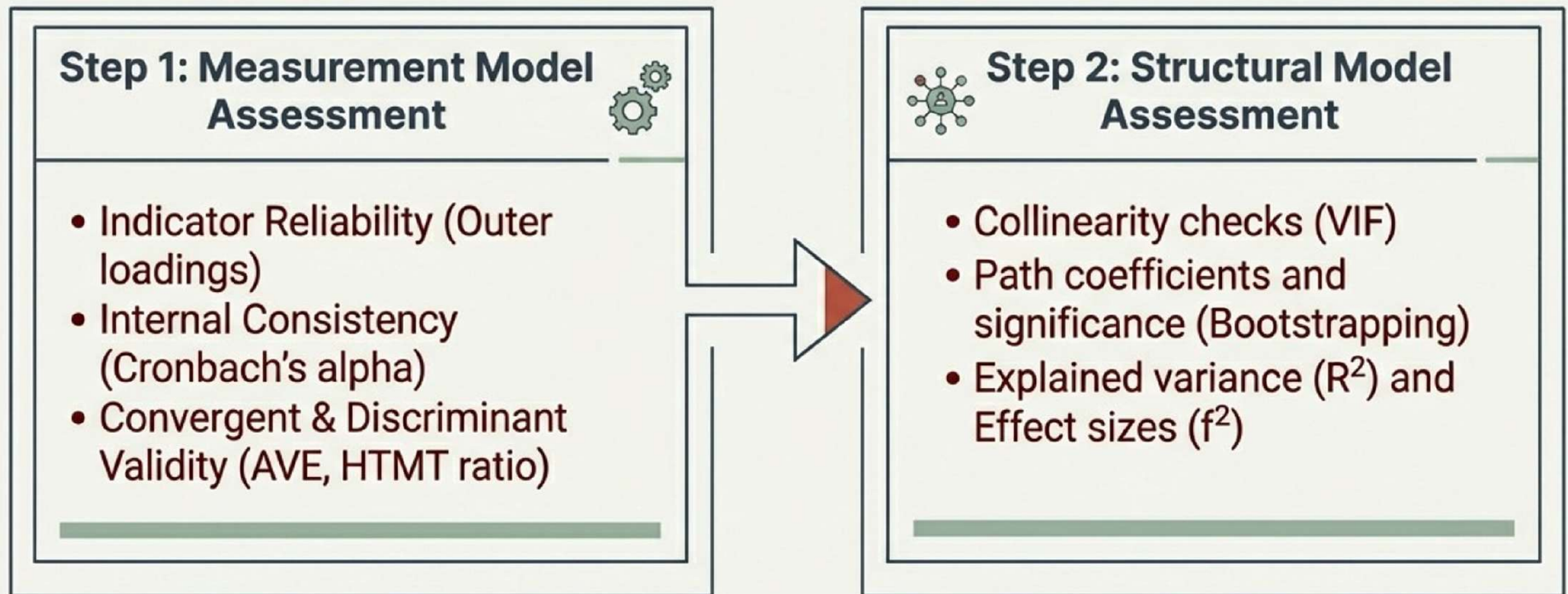
Project Execution Plan (12-Week Schedule)



Key Milestones:

- Weeks 1-5: Model refinement & Pilot testing
- Week 6: Ethics Submission
- Weeks 7-9: Sampling & Data Collection (Critical Phase)
- Week 11: PLS-SEM Analysis

Data Analysis Strategy



Goal: To statistically confirm which barriers (Trust, Habit, Usability) are the strongest predictors of discontinuance using SmartPLS 4.

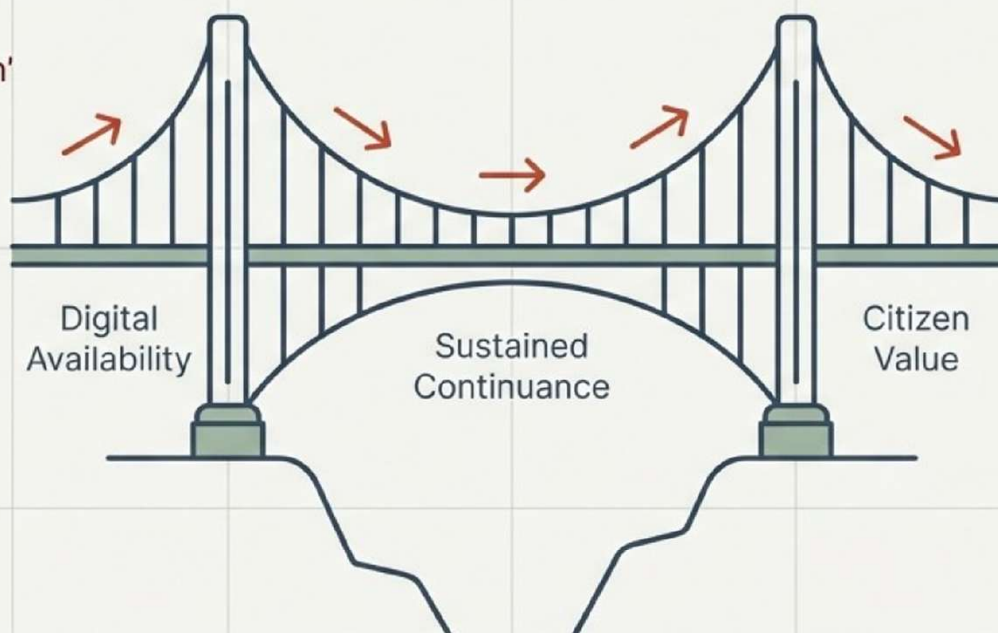
Summary & Expected Outcomes

The Goal:

Moving beyond 'adoption' to understand the drivers of sustained engagement.

The Output:

Evidence-based recommendations to help GRA bridge the gap between digital availability and citizen value.



Final Thought:

Sustainable e-government requires more than a launch; it requires a system designed for habit, trust, and continued value.

Source: Guyana Public Sector Digitalization Analysis / UN E-Government Survey 2022

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- Source: Various Academic and Policy Documents.

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Source: Various Academic and Policy Documents.

Thank You

Questions & Answers

